DSA REFERENCE

1D Dynamic Programming

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Cache the answer for each smaller one-dimensional state so overlapping subproblems are solved once.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

DSA / DYNAMIC-PROGRAMMING-1D

DSA

1D Dynamic Programming

CORE IDEA

Cache the answer for each smaller one-dimensional state so overlapping subproblems are solved once.

Recognition cues

+ count or optimize across a sequence
+ take/skip choices
+ prefix-dependent answer
+ repeated recursion state

Implementation sequence

+ write state in words
+ derive transition from the last decision
+ set base cases
+ choose iteration order
+ compress only after correctness

Invariant

dp[i] has one precise meaning and is final before any later state reads it.

Worked reasoning

House Robber uses dp[i] = max(dp[i-1], dp[i-2] + value[i]); the last decision is skip or take.

Complexity

Usually O(n) time and O(n) space, often compressible to O(1) when only recent states are read.

Failure modes

- [ ] vague state meaning
- [ ] wrong base cases
- [ ] invalid iteration order
- [ ] greedy reasoning where future effects matter

Practice ladder

1D Dynamic Programming: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Climbing Stairs**	State the cue, invariant, edge case, and proof before implementation.
2	**House Robber**	State the cue, invariant, edge case, and proof before implementation.
3	**Coin Change**	State the cue, invariant, edge case, and proof before implementation.
4	**Word Break**	State the cue, invariant, edge case, and proof before implementation.
5	**Longest Increasing Subsequence**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.